

Skills Learned and Applications in Asset Management

Two-page summary of the technical capabilities developed through the thesis project on machine-learning-based stock selection and mean-variance portfolio optimization.

1. End-to-end quantitative investment workflow

Through this project, I learned how to build a complete quantitative investment process, starting from raw market data and ending with portfolio construction, backtesting, risk evaluation and benchmark comparison. The work was not only theoretical: it required creating a practical pipeline that connects data preparation, machine learning, stock selection, optimization and portfolio monitoring.

The first skill developed was dataset construction. I learned how to define an investment universe, clean tickers, download historical prices, compute monthly returns and organize the information into a panel dataset where each observation is identified by ticker and month. This structure is essential in asset management because portfolio models depend on clean, comparable and reproducible inputs.

The second skill was feature engineering. I transformed historical prices and returns into financial variables such as momentum, volatility, trend indicators and maximum drawdown. These variables summarize different investment dimensions: recent performance, risk level, downside exposure and trend persistence. I also learned the importance of correct time alignment: variables used at time t must contain only information available at that time, while the target is the following month return.

2. Machine learning for stock selection

A central part of the project was learning how to apply supervised machine learning models to financial forecasting. Random Forest and Neural Network models were used to estimate one-month-ahead returns for each stock. However, the objective was not to predict the exact return of every security with perfect precision, because monthly equity returns are extremely noisy. The objective was to generate a ranking signal able to identify, on average, the stocks with relatively better future prospects.

This distinction is crucial in asset management. A predictive model can be valuable even when its point forecasts are imperfect, provided that it ranks securities in a useful way. For this reason, I learned to evaluate models not only through error metrics such as MAE and RMSE, but also through ranking-oriented indicators such as the Information Coefficient, which measures the relationship between predicted scores and realized future returns.

The models were tested through a monthly walk-forward validation protocol. At each date, the model was trained only on past information and then used to predict the following month. This procedure is closer to a real investment process and helps reduce look-ahead bias, overfitting and unrealistic backtest results.

3. From prediction to portfolio construction

I learned how to transform model forecasts into portfolio decisions. At each monthly rebalancing date, stocks were ranked by their predictive score and only the highest-ranked securities were selected, for example the top 20% of the universe. This turns the machine learning output into a disciplined stock selection rule that is transparent, repeatable and operational.

The selected stocks can then be combined with Markowitz mean-variance optimization and maximum-Sharpe allocation. This taught me how expected returns, volatility and covariance interact when assigning portfolio weights. It also showed the practical limits of optimization: small input errors can produce unstable weights, excessive concentration and portfolios that look good in theory but are difficult to implement. For this reason, the most realistic approach is to combine a predictive selection layer with controlled and constrained portfolio construction.

Area	What I learned	Use in asset management
Data engineering	Ticker cleaning, return calculation, panel dataset creation	Reliable inputs for quantitative research
Feature engineering	Momentum, volatility, trend and drawdown indicators	Conversion of market data into investment signals
Machine learning	Random Forest and Neural Network forecasting	Systematic stock ranking and active selection
Optimization	Markowitz and maximum-Sharpe allocation	Risk-adjusted portfolio construction

4. Backtesting, risk analysis and robustness

Another major skill developed was empirical portfolio evaluation. I learned how to backtest investment strategies over time, compare them with passive benchmarks such as the S&P 500 and NASDAQ, and evaluate whether the strategy creates value after considering risk. This is fundamental because a portfolio should not be judged only by total return. It must also be assessed through volatility, Sharpe ratio, drawdown, transaction costs and stability across market regimes.

The project also included stress testing and Monte Carlo simulation. These tools are useful because they show how a strategy may behave under adverse scenarios, how large losses can become, and how sensitive the final result is to the order of returns. In professional asset management, this type of analysis is necessary to understand whether a strategy is robust or only successful in a specific historical period.

I also learned how to interpret model behaviour through feature importance. This helps identify which variables contribute most to the predictive signal. Even when complex models are used, an asset manager must understand the economic logic behind the model, because investment decisions need to be explainable to clients, committees and risk teams.

5. Direct applications in asset management

The framework can be applied to quantitative equity research. A large universe of stocks can be screened automatically, and the model can rank securities according to their expected relative attractiveness. This allows analysts and portfolio managers to focus attention on the most promising names instead of manually reviewing the entire universe.

It can also be applied to active portfolio construction. Machine learning signals can define which stocks enter the portfolio, while optimization can determine how capital is allocated among them. In this way, the investment process combines prediction and risk control. The goal is to build a portfolio with a better risk-return profile than a passive benchmark, not simply to chase the highest possible return.

A further application is dynamic portfolio monitoring. By updating the model on a monthly basis, the manager can observe changes in rankings, risk indicators, volatility regimes and drawdown exposure. This creates a more adaptive investment process than a static buy-and-hold strategy, while still remaining systematic and rules-based.

Finally, the project is useful for performance attribution and strategy validation. By comparing machine learning portfolios with passive benchmarks, equal-weighted portfolios and Markowitz portfolios, it becomes possible to understand whether performance comes from stock selection, allocation decisions, market exposure or risk control. This is directly relevant for professional reporting and investment committee discussions.

6. Professional value of the project

Overall, this project taught me how to move from financial theory to an implementable investment process. I developed practical skills in data preparation, coding, machine learning, portfolio construction, backtesting and risk interpretation. More importantly, I learned that a model is useful in asset management only if it can be translated into a disciplined decision rule, tested out of sample, compared with relevant benchmarks and evaluated under realistic risk conditions.

The professional value of the project is the combination of three areas that are increasingly important in modern asset management: financial theory, quantitative programming and machine learning. These skills can support equity selection, portfolio optimization, active management, factor research, risk monitoring and the development of systematic investment strategies.

Reference basis: Master thesis project, *Stock Selection with Machine Learning: Integrating Predictive Models into Mean-Variance Portfolio Optimization*.